

92. Let $786 \div 24 \times x = 6.55$.

Then, $\frac{786}{24} \times x = 6.55 \Rightarrow 32.75 x = 6.55$

$\Rightarrow x = \frac{6.55}{32.75} = \frac{655}{3275} = \frac{1}{5} = 0.2$.

93. Number of pieces = $\frac{857}{12.6} = \frac{8570}{126} = 68.01 \approx 68$.

94. Quantity of blood donated in 2 years = (350×3) ml
= 1050 ml = 1.05 litres.

$\therefore$ Quantity of blood donated in 6 years

= $\left(\frac{1.05}{2} \times 6\right)$ litres = 3.15 litres.

95. Daily consumption = 1600 ml = 1.6 litres.

Consumption in 4 weeks = $(1.6 \times 4 \times 7)$ litres = 44.8 litres.

96. Length of each piece = $\left(\frac{1}{8}\right)$ m = 0.125 m.

$\therefore$ Required number of pieces

= $\left(\frac{37.5}{0.125}\right) = \left(\frac{375 \times 100}{125}\right) = 300$.

97. Suppose commodity X will cost 40 paise more than Y after z years. Then,

$(4.20 + 0.40z) - (6.30 + 0.15z) = 0.40$

$\Leftrightarrow 0.25z = 0.40 + 2.10 \Leftrightarrow z = \frac{2.50}{0.25} = \frac{250}{25} = 10$.

$\therefore$ X will cost 40 paise more than Y 10 years after 2004 i.e., in 2014.

98. Let the number of 10P coins be x .

Then, number of 5P coins = $(80 - x)$.

$\therefore 0.1x + 0.05(80 - x) = 6.25$

$\Rightarrow 0.1x + 4 - 0.05x = 6.25 \Rightarrow 0.05x = 2.25$

$\Rightarrow x = \frac{2.25}{0.05} = \frac{225}{5} = 45$.

Hence, number of 10P coins = 45. And, number of 5P coins = $(80 - 45) = 35$.

99. $0.121212\dots = 0.\overline{12} = \frac{12}{99} = \frac{4}{33}$.

100. $0.125125\dots = 0.\overline{125} = \frac{125}{999}$.

101. $0.\overline{47} = \frac{47}{99}$.

102. $0.\overline{36} = \frac{36}{99} = \frac{4}{11}$.

103. $1 \div 0.2 = \frac{1}{0.2} = \frac{10}{2} = 5$; $0.\overline{2} = 0.222\dots$; $(0.2)^2 = 0.04$.

$0.04 < 0.2 < 0.22\dots < 5$.

Since 0.04 is the least, so $(0.2)^2$ is the least.

104. $1.\overline{27} = 1 + 0.\overline{27} = 1 + \frac{27}{99} = 1 + \frac{3}{11} = \frac{11+3}{11} = \frac{14}{11}$.

105. $0.\overline{423} = \frac{423-4}{990} = \frac{419}{990}$.

106. $0.29\overline{56} = \frac{2956-29}{9900} = \frac{2927}{9900}$.

107. $0.84\overline{181} = \frac{84181-841}{99000} = \frac{83340}{99000} = \frac{463}{550}$.

$\therefore$ Required difference = $(550 - 463) = 87$.

108. $4.\overline{12} = 4 + 0.\overline{12} = 4 + \frac{12-1}{90} = 4 + \frac{11}{90}$.

109. $2.\overline{8768} = 2 + 0.\overline{8768} = 2 + \frac{8768-8}{9990} = 2 + \frac{8760}{9990} = 2 + \frac{292}{333}$.

110. $0.\overline{2} + 0.\overline{3} + 0.\overline{32} = \left(\frac{2}{9} + \frac{3}{9} + \frac{32}{99}\right) = \left(\frac{22+33+32}{99}\right) = \frac{87}{99} = 0.\overline{87}$.

111. $0.\overline{142857} + 0.\overline{285714} = \frac{142857}{999999} + \frac{285714}{999999}$
= $\left(\frac{142857}{999999} \times \frac{999999}{285714}\right) = \frac{1}{2}$.

112. $3.\overline{87} - 2.\overline{59} = (3 + 0.\overline{87}) - (2 + 0.\overline{59}) = \left(3 + \frac{87}{99}\right) - \left(2 + \frac{59}{99}\right)$
= $1 + \left(\frac{87}{99} - \frac{59}{99}\right) = 1 + \frac{28}{99} = 1.\overline{28}$.

113. $3.\overline{36} - 2.\overline{05} + 1.\overline{33} = [(3 + 0.\overline{36}) + (1 + 0.\overline{33})] - (2 + 0.\overline{05})$
= $\left[4 + \left(\frac{36}{99} + \frac{33}{99}\right)\right] - \left[2 + \frac{5}{99}\right]$
= $2 + \left(\frac{36}{99} + \frac{33}{99} - \frac{5}{99}\right) = 2 + \frac{64}{99} = 2.\overline{64}$.

114. $0.\overline{09} \times 7.\overline{3} = \frac{9}{99} \times 7\frac{3}{9} = \frac{1}{11} \times \frac{66}{9} = \frac{2}{3} = 0.\overline{6}$.

115. $0.34\overline{67} + 0.13\overline{33} = \frac{3467-34}{9900} + \frac{1333-13}{9900} = \frac{3433+1320}{9900}$
= $\frac{4753}{9900} = \frac{4801-48}{9900} = 0.48\overline{01}$.

116. $(8.\overline{31} + 0.\overline{6} + 0.00\overline{2}) = 8 + \frac{31-3}{90} + \frac{6}{9} + \frac{2}{900}$
= $\frac{7200+280+600+2}{900}$
= $\frac{8082}{900} = 8\frac{882}{900} = 8 + \frac{979-97}{900} = 8.97\overline{9}$.

117. $\overline{2.75} + \overline{3.78} = (-2 + 0.75) + (-3 + 0.78)$
= $-5 + (0.75 + 0.78) = -5 + 1.53$
= $-5 + 1 + 0.53 = -4 + 0.53 = \overline{4.53}$.

118. $\frac{547527}{82} = \frac{54.7527}{0.0082} = \left(\frac{547.527}{0.0082} \times \frac{1}{10}\right) = \frac{x}{10}$.

119. $\frac{29.94}{1.45} = \frac{299.4}{14.5} = \left(\frac{2994}{14.5} \times \frac{1}{10}\right) = \frac{172}{10} = 17.2$.

120. $1.6 \times 21.3 = \left(\frac{16}{10} \times \frac{213}{10}\right) = \left(\frac{16 \times 213}{100}\right) = \frac{3408}{100} = 34.08$.

121. $\frac{1}{0.0006198} = \frac{10000}{6.198} = \left(10000 \times \frac{1}{6.198}\right)$
= $(10000 \times 0.16134) = 1613.4$.